

ISRO SPACE APPLICATIONS CENTRE (SAC) — AHMEDABAD

SPACEFLIGHT MICROELECTRONICS QUALIFICATION CERTIFICATE
adhering to MIL-STD-883 Method 1015 | MIL-PRF-38535 Class V | ESA ECSS-Q-ST-60C

1. Component Identity & Environmental Test Parameters

Component Serial ID:	ISRO-SAC-QUAL-2026-0042	Test Chamber Temp:	125.0°C (HTOL)
Device Family:	DIGITAL_IC	Vacuum Pressure:	10 Torr (TVAC)
Project Flight Code:	ISRO_GAGANYAAN_SAC_LOT_01	ATE Station ID:	SAC-AHMEDABAD-ATE-01
Screening Parameter:	Quiescent Current (µA)	Specification USL:	45.0 µA

2. Telemetry Measurements & 95% Conformal Uncertainty Interval

Measurement Checkpoint	Parametric Value	Robust Z-Score	Conformal 95% Bounds
0h Baseline Measurement	11.20 µA	0.00 σ	N/A (Pre-Stress Base)
24h Chamber Telemetry	12.10 µA	0.45 σ	Live Telemetry Sensor
168h Forecast Trajectory	14.80 µA	0.81 σ	[13.50 - 16.10] µA

3. Game-Theoretic SHAP Physics Feature Attribution Breakdown

SHAP (Shapley Additive exPlanations) uses cooperative game theory to quantify the exact marginal contribution of each physical stressor and parametric variable to the forecasted 168-hour degradation trajectory:

Physical Feature / Kinetic Variable	Observed Value	SHAP Attribution Value	Governing Physics Failure Mechanism
24h Current Measurement (X_24h)	12.10 µA	+0.715 µA	Arrhenius Thermal Leakage Base
Kinetic Degradation Velocity (v24)	0.0375 µA/hr	+0.469 µA	Black's Electromigration / Oxide Drift
Robust Population Z-Score (Z_mad)	0.45 σ	+0.189 µA	Wafer Periphery Process Gradient
Arrhenius Activation Factor (Ea=0.68 eV)	125.0°C Stress	+0.250 µA	Shockley-Read-Hall (SRH) Trap Gen

4. Quality Assurance (QA) Engineer Decision Audit Report

■ QUALIFIED FOR 24H FLIGHT RELEASE (AUTO-PASS)
QA INSPECTION SUMMARY — NOMINAL DEGRADATION: Component ISRO-SAC-QUAL-2026-0042 exhibits nominal Arrhenius thermal degradation kinetics ($v_{24} = 0.0375 \text{ µA/hr}$). The non-parametric 95% Conformal Prediction Upper Bound (16.10 µA) is safely below the Specification USL (45.0 µA) with a 64.2% safety margin . QA Recommendation: Approve immediate 24-hour early release from burn-in chamber. Safe for spaceflight integration.

5. ISRO QA Inspector Sign-Off & Cryptographic Verification

Lead QA Reliability Engineer:	_____	Date & Stamp:	_____
ISRO SAC Qualification Authority:	_____	SHA-256 Audit Verification:	a7f9b2c4e18d3091...